DOE ASPECT · DE-FOA-0003647 · TOPIC AREA 1, SUBTOPIC 1a

Bibliography and References Cited

Supporting material. It fixes the boundary between prior art and the PlastiBioFuel contribution, sources the market-scale statement in the concept paper, and shows the subaward partner's published record in enzyme immobilization.

6	3	1
PEER-REVIEWED SOURCES	WITH DR. MA AS AUTHOR	PROVISIONAL PATENT

ENZYMATIC DEPOLYMERIZATION OF PET

1 Lu, H.; Diaz, D. J.; Czarnecki, N. J.; Zhu, C.; Kim, W.; Shroff, R.; Acosta, D. J.; Alexander, B. R.; Cole, H. O.; Zhang, Y.; Lynd, N. A.; Ellington, A. D.; Alper, H. S. **Machine learning-aided engineering of hydrolases for PET depolymerization.** *Nature* 2022, 604 (7907), 662-667. DOI 10.1038/s41586-022-04599-z · PMID 35478237

> Source of FAST-PETase and of the machine-learning method that produced it, from the McKetta Department of Chemical Engineering at the University of Texas at Austin. PlastiBioFuel did not develop this enzyme and makes no claim to it. Cited to fix the boundary between prior art and the PlastiBioFuel contribution, and because the paper reports depolymerization of untreated post-consumer PET from 51 thermoformed products and of a commercial water bottle, which is the feedstock class this project uses.

2 Tournier, V.; Topham, C. M.; Gilles, A.; David, B.; Folgoas, C.; Moya-Leclair, E.; Kamionka, E.; Desrousseaux, M.-L.; Texier, H.; Gavalda, S.; Cot, M.; Guemard, E.; Dalibey, M.; Nomme, J.; Cioci, G.; Barbe, S.; Chateau, M.; Andre, I.; Duquesne, S.; Marty, A. **An engineered PET depolymerase to break down and recycle plastic bottles.** *Nature* 2020, 580 (7802), 216-219. DOI 10.1038/s41586-020-2149-4 · PMID 32269349

> Source of the market-scale statement in section 1 of the concept paper: the paper reports PET as the most abundant polyester plastic with almost 70 million tons manufactured annually worldwide. Terephthalic acid is its principal monomer, which places the target chemical far above this NOFO's one million metric ton threshold. The paper is also the productivity benchmark for enzymatic PET depolymerization and establishes the competing route that returns monomer to virgin-equivalent resin.

3 Yoshida, S.; Hiraga, K.; Takehana, T.; Taniguchi, I.; Yamaji, H.; Maeda, Y.; Toyohara, K.; Miyamoto, K.; Kimura, Y.; Oda, K. **A bacterium that degrades and assimilates poly(ethylene terephthalate).** *Science* 2016, 351 (6278), 1196-1199.
DOI 10.1126/science.aad6359 · PMID 26965627

> Origin of the PETase and MHETase system. Establishes that the enzymatic route converts PET to terephthalic acid and ethylene glycol, which is why terephthalic acid is the primary product this project targets.

ENZYME IMMOBILIZATION IN ORGANIC AND METAL-ORGANIC FRAMEWORKS

4 Wang, X.; Lan, P. C.; **Ma, S.** **Metal-Organic Frameworks for Enzyme Immobilization: Beyond Host Matrix Materials.** *ACS Central Science* 2020, 6 (9), 1497-1506.
DOI 10.1021/acscentsci.0c00687 · PMID 32999925

> Written from the Department of Chemistry at the University of North Texas. Dr. Shengqian Ma leads framework and immobilization chemistry on the PlastiBioFuel subaward under Sponsored Research Agreement SRA1016.

5 Li, M.; Qiao, S.; Zheng, Y.; Andaloussi, Y. H.; Li, X.; Zhang, Z.; Li, A.; Cheng, P.; **Ma, S.**; Chen, Y. **Fabricating Covalent Organic Framework Capsules with Commodious Microenvironment for Enzymes.** *Journal of the American Chemical Society* 2020, 142 (14), 6675-6681.
DOI 10.1021/jacs.0c00285 · PMID 32197569

> Covalent organic framework capsules as enzyme hosts. The paper reports improved conformational freedom and enhanced mass transfer as the mechanism behind retained activity, which is the same variable this project proposes to measure and optimize at bench scale.

6 Zhang, S.; Zheng, Y.; An, H.; Aguila, B.; Yang, C.-X.; Dong, Y.; Xie, W.; Cheng, P.; Zhang, Z.; Chen, Y.; **Ma, S.** **Covalent Organic Frameworks with Chirality Enriched by Biomolecules for Efficient Chiral Separation.** *Angewandte Chemie International Edition* 2018, 57 (51), 16754-16759.
DOI 10.1002/anie.201810571 · PMID 30359485

> Covalent immobilization of enzymes and other biomolecules into covalent organic frameworks with function retained, from the same group.

PLASTIBIOFUEL INTELLECTUAL PROPERTY

7 Ward, M. D. **U.S. Provisional Patent Application 63/848,456**, filed July 2025.

> Covers covalent organic framework immobilization, the spiral-flow reactor geometry, and the integrated depolymerization process. Conversion is being settled with patent counsel.

VERIFICATION

> References 1 through 6 were retrieved from PubMed and verified by PMID on October 5, 2026. Titles, journals, volumes, issues, page ranges, and DOIs match the PubMed records. Reference 2's market figure is quoted from that paper's own abstract.